

Geography Ch-16 (Must-Do): The Tropical Monsoon & Tropical Marine Climate (Am)

Tier 1: Must-Do | Source: GC Leong + Majid Husain (Indian & World Geography)
Grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

The Tropical Monsoon Climate (Am): Seasonal Wind Reversal, Onset, Burst & Retreat

GS-I (Physical Geography - Climatology)
Geography

■ NEWS TRIGGER

Foundation concept (GC Leong; Majid Husain). Tropical monsoon climate lies 10-30 deg N & S on the eastern/southern margins of continents; its defining feature is the COMPLETE SEASONAL REVERSAL of wind direction. Best developed in the Indian subcontinent.

■ INTRO & ORIGIN

The monsoon climate is a modification of the trade-wind belt: winds reverse with the seasons because of differential heating of land and sea. Summer brings onshore, rain-bearing SW winds; winter brings offshore, dry NE winds. The whole year's rain is concentrated in the summer months, giving a strongly seasonal rainfall regime.

Driven by differential heating of land and sea + the seasonal migration of the ITCZ (pressure belts shift ~5 deg with the overhead sun). In summer, intense low pressure over heated Asia draws in moist ocean air; in winter, high pressure over the cold continent pushes dry air seaward. GC Leong: reversal is 'the key-note of monsoon climate' (Conrad).

■ KEY DATA

Season	Wind	Weather
Hot dry season (Mar-May)	Land-heating, pre-monsoon	High temp; Loo; local storms (Kalbaisakhi)
SW Monsoon (Jun-Sep)	Onshore SW winds (onset & burst)	Heavy convectional + orographic rain
Retreating Monsoon (Oct-Nov)	Weakening, reversing winds	Tropical cyclones (Bay of Bengal)
Cool dry season (Dec-Feb)	Offshore NE winds	Dry; NE monsoon rain on TN coast

■ STATIC THEORY

- **Three-season model (GC Leong):** (1) hot, dry season (Mar-May) with rising temperature and local storms; (2) the rainy SW-monsoon season (Jun-Sep) with the 'onset' and 'burst' of monsoon; (3) the retreating/cool season (Oct-Feb).
- **Onset & burst:** the sudden arrival of the SW monsoon (over Kerala ~1 June) with violent thunder, lightning and torrential rain = the 'burst' of the monsoon.
- **Rainfall regime:** highly seasonal - almost the entire annual total falls in the 3-4 summer months; rain is convectional + orographic; travelling depressions along the ITCZ add rain moving northward with the season.
- **Tropical Marine climate (variant):** on trade-wind coasts (e.g. eastern Central America, Caribbean, eastern Brazil, Madagascar's east coast), constant onshore Trade Winds give abundant rain (steadier than monsoon, ~150 cm+), moderated by the sea - more reliable and evenly spread than the monsoon.
- **Retreating monsoon:** as winds reverse in Oct-Nov, the Bay of Bengal spawns the most severe tropical cyclones of the year (D.R. Khullar).

■ MUST-KNOW FACTS

1. Koppen Am; located 10-30 deg N & S, eastern/southern continental margins; best developed in India.
2. Defining feature = complete SEASONAL REVERSAL of wind direction (Conrad's 'key-note').

■ UPSC TRAPS

- ✗ The tropical monsoon and tropical marine climates are identical.
- ✓ Both are trade-wind based, but monsoon rain is highly SEASONAL (summer-concentrated) while tropical-marine rain is steadier and more evenly distributed through the year.

3. SW monsoon 'burst' over Kerala ~1 June; whole year's rain concentrated in summer.
4. Tropical Marine = trade-wind coasts, steadier & more reliable rain than monsoon, sea-moderated.
5. Retreating monsoon (Oct-Nov) = peak tropical-cyclone season in the Bay of Bengal.

- ✗ The monsoon is simply a large sea breeze caused only by land-sea heating.
- ✓ Land-sea heating is one driver, but the seasonal migration of the ITCZ/pressure belts and jet streams are equally essential to the mechanism.
- ✗ Monsoon 'burst' means the end of the rainy season.
- ✓ The 'burst' is the sudden ONSET of heavy monsoon rain (early June), not its end.

■ MAINS ANGLE

Explain the mechanism of the tropical monsoon climate and distinguish it from the tropical marine climate, highlighting why the monsoon's seasonality makes it both a lifeline and a hazard.

■ STUDY LINK

Geography -> World Climatic Types -> Tropical Monsoon (Am) & Tropical Marine; core of the Indian climate

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: India's 2026 Below-Normal SW Monsoon Forecast (90% of LPA)

GS-I (Climate of India) + GS-III (Agriculture / Economy)
Geography - Current Affairs

■ NEWS TRIGGER

2026: IMD forecast a BELOW-NORMAL southwest monsoon at ~90% of LPA (revised down from 92%) - the first below-normal monsoon since 2023 - citing developing El Nino; onset over Kerala slipped to the first week of June; a positive IOD was expected late-season to offer partial relief.

■ INTRO & ORIGIN

The 2026 monsoon forecast is the living application of the monsoon-climate chapter: it shows how the seasonal SW-monsoon system is modulated by ENSO (El Nino) and the Indian Ocean Dipole (IOD), and why a 'below-normal' label carries huge agricultural and economic stakes.

IMD's Long Range Forecast expresses seasonal rain as a % of the LPA (1971-2020 average, ~87 cm / 868.6 mm). El Nino suppresses monsoon rain; a positive IOD can partly offset it.

■ KEY DATA

Parameter	2026 value	Meaning
Forecast	~90% of LPA (below normal)	First below-normal since 2023
LPA (1971-2020)	~87 cm (868.6 mm)	Benchmark for 'normal'
ENSO	El Nino developing	Suppresses monsoon rainfall
IOD	Neutral -> positive late season	Positive IOD can boost late rain
Onset (Kerala)	First week of June (delayed)	Slipped from ~26 May forecast

■ STATIC THEORY

- **LPA (Long Period Average):** the 50-year mean monsoon rainfall (currently 1971-2020, ~87 cm). IMD categories: Normal 96-104%; Below Normal 90-95%; Deficient < 90%; Above Normal 105-110%; Excess > 110%.

- **El Nino (ENSO warm phase):** abnormal warming of the central-east Pacific weakens the Walker circulation and typically **SUPPRESSES** the Indian SW monsoon (cf. drought years 2009, 2015).
- **Indian Ocean Dipole (IOD):** a see-saw of SST between the western and eastern Indian Ocean. A **POSITIVE** IOD (warm west) enhances Indian monsoon rain and can partly offset El Nino; negative IOD does the opposite.

■ MUST-KNOW FACTS

1. IMD 2026 SW-monsoon forecast: ~90% of LPA = 'below normal'; first below-normal since 2023.
2. LPA (1971-2020) ~ 87 cm (868.6 mm); Normal band = 96-104% of LPA.
3. El Nino suppresses the monsoon; positive IOD can partially offset it.
4. 2026 monsoon onset over Kerala delayed to the first week of June.
5. Below-normal monsoon threatens rain-fed Kharif crops (pulses, oilseeds) and rural demand.

■ UPSC TRAPS

- ✗ 'Below normal' and 'deficient' are the same IMD category.
- ✓ Below Normal = 90-95% of LPA; Deficient = below 90% - distinct IMD categories.
- ✗ A positive IOD worsens the monsoon.
- ✓ A **POSITIVE** IOD generally **ENHANCES** Indian monsoon rainfall and can offset El Nino; the **NEGATIVE** phase suppresses it.

■ MAINS ANGLE

Analyse how *ENSO (El Nino)* and the *IOD* jointly modulate the *Indian monsoon*, using the *2026 below-normal forecast* to discuss risks to *agriculture, water and the rural economy*.

■ STUDY LINK

Geography -> Indian Monsoon + GS-III agriculture/economy; connect to ENSO/IOD teleconnections (Ch-12/13 Advanced)